CAT P. LE

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SUMMARY

Machine learning scientist with 8+ years of experience developing and deploying AI solutions in computer vision, natural language processing, and large-language models. Skilled in data analysis, image generation, and time-series forecasting, with proven success delivering scalable, high-impact AI systems through cross-functional collaboration.

Education

DUKE UNIVERSITY Durham, NC
Ph.D., Electrical and Computer Engineering May 2023
Thesis: *Task Affinity and Its Applications in Machine Learning*
Advisor: Dr. Vahid Tarokh

CALIFORNIA INSTITUTE OF TECHNOLOGY Pasadena, CA
M.S., Electrical Engineering, GPA: 4.00 Jun 2017
Advisor: Dr. Babak Hassibi

RUTGERS UNIVERSITY New Brunswick, NJ
B.S., Electrical and Computer Engineering, GPA: 4.00 Jun 2017
Honors: Summa Cum Laude, Matthew Leydt Society, John B. Smith Award, Tau Beta Pi

Professional Experience

UL SOLUTIONS – AI Safety Scientist Oct 2025 – Present

- Develop an internal LLM-powered evaluation tool to streamline and accelerate product assessments against ISO/IEC standards, significantly improving efficiency and consistency in compliance workflows.
- Contribute to ISO/IEC standards alignment and implementation, while supporting external clients through advisory services and product testing to ensure regulatory compliance and safety readiness.

RAYTHEON & DUKE UNIVERSITY – Postdoctoral Scientist May 2023 – Oct 2025

- Develop a flow-based generative model for backscatter imaging of titanium alloys.
- Design surface-based detector using Transformer, Attentional Copulas for mechanical malfunctions.

AMAZON - Research Scientist Jun 2022 – Sep 2022

- Develop open-domain dialogs evaluation systems with BERT, LSTM, and causal inference analysis.
- Help improve the prediction performance for the customer's and expert's ratings.

MOTOROLA SOLUTIONS - Software Engineer Jun 2017 – Aug 2018

- Develop a Camera Shutter Synchronization System with LED Strobing for license plate recognition cameras.
- Improve energy consumption and the performance of the license plate and facial recognition cameras.

Featured Publications

Generative Models and Bootstrapping for Titanium Alloy Microstructure Analysis UAI 2025
Cat P. Le, M. LaRosa, Vahid Tarokh

Improving Open-Domain Dialog Evaluation with a Counterfactual LSTM IWSDS 2023
Cat P. Le, Luke Dai, Michael Johnston, Yang Liu, Marilyn Walker, Reza Ghanadan Best Paper Award

Task Affinity with Maximum Bipartite Matching in Few-Shot Learning ICLR 2022
Cat P. Le, Juncheng Dong, Mohammadreza Soltani, Vahid Tarokh

Skills

Technical: Pytorch, Python, Numpy, Scipy, Scikit-learn, Pandas, Matplotlib, OpenCV, Tensorflow, Keras, C++, Matlab
Research Area: Neural Network Architecture, Learning Algorithm, Computer Vision, Large-Language Model
ML Framework: Transfer Learning, Continual Learning, Few-Shot Learning, Prompt Tuning, Reinforcement Learning